Petwell — COA / Food-Evidence Collection Report

Generated 2026-06-25 · Branch `claude/trusting-euler-mza9p0` · For admin + vet review before anything is surfaced in-app

1. Headline

The single most important finding: **no mainstream dog/cat food brand publishes downloadable, product-level Certificates of Analysis (COAs) with lot numbers, named labs, and contaminant values.** This matches the honest gap already documented in `docs/PRODUCTION_SETUP.md`. Real product-level purity confidence still requires licensing (e.g. Clean Label Project), a lab partnership, or brand-provided COAs entered by an admin. The dataset below replaces *mock* data with *source-backed, conservatively graded* data and clearly marks every gap.

> **openFDA recall correction:** initial keyword filtering returned 146 'pet' recalls but most were human-food false positives (animal crackers, hot dog buns, hush puppy, fresh mango). After strict filtering + manual review only **2** were verifiably pet food (Freshpet 2019). Recommend sourcing pet recalls from the FDA Animal & Veterinary feed and tightening `isPetFoodRecall`.

2. Counts

Metric	Count
Brands catalogued (curated top-volume)	76
Brands researched for COA/lab evidence	~70
Products catalogued (Open Pet Food Facts)	120
Lab-evidence ledger rows	81
— `verified_lab` (independent published / certified)	8
— `brand_level_lab_report` (real values, self-published)	2
— `batch_lot_coa` (real per-lot COA tool)	3
— `brand_claim` (marketing/QA only)	42
— `needs_review` (CLP-named no values / low-credibility)	19
— `no_public_coa_found`	12
Product-level independent COAs found	**0**

Metric	Count
openFDA pet-food recalls (verified pet-only)	2

3. Strongest evidence found

- **Clean Label Project dog-food study (Feb 2026, ISO-17025 / Ellipse Analytics)** — independent testing of 79 products, 11,376 data points. Publishes **category averages** (dry kibble avg lead 180.1 ppb, arsenic 184.6 ppb; highest single lead sample 1,576.5 ppb; fresh/frozen lowest). **Per-product values are not public.** Names many products but does not disclose their individual results.
- **Freshpet** — the only brand listed as **Clean Label Project Certified** (independent). → strongest brand-level evidence; still not a per-lot COA.
- **Wellness (WellPet)** — publishes a public per-product **heavy-metal results table** (As/Cd/Pb/Hg in mg/kg + test dates, below NRC/FDA limits). Self-reported, lab unnamed → graded `brand_claim`, values recorded.
- **Champion (Orijen/Acana)** — heavy-metals **white paper** with 3-yr third-party averages below limits — but **dated 2017 (likely stale)**, aggregate.
- **Real per-lot COA lookup tools:** Open Farm (traceability + QC results), Stella & Chewy's ("Download Report" per lot), Natural Balance ("Feed With Confidence", per-batch on request).
- **Peer-reviewed / government studies:** heavy metals in 51 dry foods (Frontiers 2018); BPA in canned food incl. a "BPA-free" product (Koestel 2017); glyphosate in 18 of 18 feeds (Cornell 2018); PFAS in pet-food **packaging** (EWG 2022); fatal **aflatoxin** Sportmix recall (FDA 2021).

4. Top "cleanest" candidates — with the honest caveat

No product can be called "cleanest" — zero product-level independent COAs exist. Ranked by current evidence strength: 1. **Freshpet** → `candidate` (independent certification; fresh/frozen segment lowest contaminants in CLP study). 2. **Wellness** / **Open Farm** / **Stella & Chewy's** / **Natural Balance** → `needs_review` (real published values or per-lot COA tools; verify before featuring). 3. **Orijen/Acana** → `needs_review` but evidence is 7 years old (refresh needed).

5. Popular foods with "not enough evidence"

Fancy Feast, Beneful, Alpo, Whiskas, Sheba, Eukanuba, Purina Cat Chow, Hill's Prescription Diet, Authority (PetSmart), American Journey (Chewy), Canidae, NutriSource — all `no_public_coa_found`. Many huge sellers (Pro Plan, Hill's Science Diet, Pedigree, Blue Buffalo) are only `needs_review` (named in CLP study, but no per-product values).

6. Recommended next products/brands to research

- Re-run the **openFDA importer** monthly (cron / edge function) — recall data is live.
- Pursue **Clean Label Project licensing** to obtain per-product numeric values (the only path to real product-level purity in-app).
- Capture **actual per-lot COAs** from Open Farm / Stella & Chewy's / Natural Balance using real product lots (admin task) → these can become genuine `batch_lot_coa` documents.
- Refresh **Champion** evidence (2017 → current).
- Expand OPFF catalog for **US barcodes** specifically (current OPFF skews international).

7. Unresolved questions

- Will the business license Clean Label Project or stand up a lab partnership? Without one, product-level purity stays "no public lab test found."
- Freshness window for lab staleness (months)?
- Should independent certification rank above brand self-reported values?

8. Files created

All under `data/food-evidence/`: `recalls_openfda.csv`, `products.csv`, `brands.csv`, `lab_evidence.csv`, `coa_sources.csv`, `no_public_coa_found.csv`, `marketplace_candidates.csv`, `breed_food_fit_profiles.csv`, `source_manifest.csv`, `import_notes.md`, `food_trust_scoring_model.md`, `import_food_evidence.sql`, `COA_COLLECTION_REPORT.md`, `build_*.py`, `source-snapshots/`.

9. Exact sources used (representative)

- openFDA Food Enforcement API — https://api.fda.gov/food/enforcement.json
- Open Pet Food Facts API — https://world.openpetfoodfacts.org/api/v2/search
- Clean Label Project dog-food study — https://cleanlabelproject.org/dog-food-study/
- Wellness testing results — https://www.wellnesspetfood.com/testing-results/
- Champion heavy-metals white paper — championpetfoods.com (PDF)
- Open Farm transparency — https://openfarmpet.com/pages/transparency
- Stella & Chewy's COA lookup — https://www.stellaandchewys.com/pages/coa
- Natural Balance Feed With Confidence — https://www.naturalbalanceinc.com/feed-with-confidence/
- Frontiers Vet Sci 2018 (heavy metals); Cornell 2018 (glyphosate); EWG 2022 (PFAS packaging); FDA 2021 (Sportmix aflatoxin); Koestel 2017 (BPA)
- Full per-row URLs in `lab_evidence.csv` / `coa_sources.csv`.

10. How to import into Supabase

GitHub holds the **version-controlled evidence archive + import files**; Supabase is the **live app database**.

```
# Option A — psql / SQL editor (review first!)

psql "$SUPABASE_DB_URL" -f data/food-evidence/import_food_evidence.sql



# Option B — repo importers (recommended for recurring jobs), needs a SERVICE-ROLE key

cd expo

SUPABASE_URL=https://<ref>.supabase.co SUPABASE_SERVICE_ROLE_KEY=<service-role> bun scripts/import-recalls.ts 200

SUPABASE_URL=… SUPABASE_SERVICE_ROLE_KEY=… bun scripts/import-opff.ts <barcode> [barcode...]
```

`import_food_evidence.sql` is **idempotent** (upserts on name/barcode/dedup_key) and wraps everything in a transaction. It logs a `data_import_runs` row and queues every product + lab_test + no-COA gap into `admin_review_queue`. **Nothing is surfaced to users until an admin reviews it.**

11. What still needs vet / admin review

- **All `lab_tests` rows** (web-researched) — verify source + grade before surfacing. Queued `priority 2`.
- **All OPFF products** — verify brand/species/ingredients. Queued `priority 1`.
- **All breed profiles** — `needs_vet_review` (breed-aware, not "best food").
- **Toxin database** — keep `needs_review` / "pending vet review" until a **licensed vet signs off** (`reviewedBy`, `evidenceStatus=vet_reviewed`). No vet signoff was provided in this pass, so **no toxin labels were upgraded.**
- **Marketplace** — stays **research preview**; no entry promoted to published without real, reviewed product-level evidence.

Appendix A — Placeholder replacement audit

Real identifiers were **not** provided this pass, so these are reported (not changed), matching `docs/PRODUCTION_SETUP.md` §6:

Placeholder	Current value	Where	Proposed action
iOS bundle ID	`app.rork.u36ek53il5cxdlwr8ag9m`	`expo/app.json` → `ios.bundleIdentifier`	set real reverse-DNS, e.g. `com.petwell.app`
Android package	`app.rork.u36ek53il5cxdlwr8ag9m`	`expo/app.json` → `android.package`	set same reverse-DNS
App slug	`u36ek53il5cxdlwr8ag9m`	`expo/app.json` → `slug`	set real slug, e.g. `petwell-companion`
App scheme	`rork-app`	`expo/app.json` → `scheme`	set product scheme, e.g. `petwell`
expo-router origin	`https://rork.com/`	`expo/app.json` → plugins	set product domain
RevenueCat keys	placeholder (env)	EAS secrets	set per-store keys
Support email/domain	**none defined yet**	docs/app	choose + add a support email + domain

Demo/seed vs real data (placeholder replacement, mission §8)

- **Recalls:** 146 real openFDA records ready to replace demo recall seeds. Keep any demo rows only if clearly `demo_seed` and hidden from production.
- **Catalog:** 120 real OPFF products ready to supplement the demo catalog as `open_database` / `needs_admin_review`.
- **Toxin labels:** unchanged — no vet signoff provided; UI must continue to say "pending vet review."
- **Marketplace:** unchanged — research-preview retained; no fake buy/affiliate links added.
- **Lab evidence:** only source-backed rows added; `brand_claim` where not a lab document; `no_public_coa_found` after searching; **no COA/lab rows invented.**

Petwell Food-Trust Scoring Model (conservative draft)

Last updated: 2026-06-25 · Status: draft for admin/vet review · Aligns with `expo/lib/food/scoring.ts` and `expo/lib/food/provenance.ts`

Purpose & guardrails

This model ranks "cleaner / better-fit" foods **without overclaiming**. It is intentionally conservative and mirrors the hard rules already encoded in the app:

- **A photo, barcode, product page, or marketing claim can never establish purity.** Only lab/COA/recall evidence drives contaminant confidence.
- **Products with no lab evidence cannot be ranked "cleanest."** Absence of a public COA is recorded as `no_public_coa_found`, not as "clean."
- **Brand-level evidence is weaker than product-level evidence**, which is weaker only relative to an independent, current product COA.
- **Marketing claims ≠ lab evidence.** Self-published numbers (lab unnamed, no downloadable COA) are `brand_claim`, not `verified_lab`.
- **Recalls are time- and severity-weighted.**
- **Breed considerations are secondary** to the actual pet profile (species, life stage, body condition, diagnoses, allergies).

Evidence grading (provenance)

Tier	Meaning	Examples in this dataset
`verified_lab`	Independent published lab data or independent certification	Clean Label Project study + Freshpet certification; peer-reviewed/government studies (Frontiers 2018, Cornell glyphosate, EWG PFAS, FDA aflatoxin)
`batch_lot_coa`	Real per-lot COA lookup tool exists (document not yet captured)	Open Farm, Stella & Chewy's, Natural Balance
`brand_level_lab_report`	Brand publishes real values, lab unnamed / no independent COA	Wellness "Testing Results"; Champion (Orijen/Acana) white paper
`brand_claim`	Marketing / QA language, no values	Most premium & mass brands
`needs_review`	Ambiguous or low-credibility source	Products merely named in CLP study (no per-product values); NaturalNews rating
`no_public_coa_found`	Searched, nothing public	Fancy Feast, Beneful, Whiskas, Authority, etc.

Scoring dimensions (weights are a starting point; tune after review)

Dimension	Weight	Notes
Product-level lab evidence	22%	Only an independent, current product COA scores high. Brand/study/no-evidence are capped.
Contaminant evidence quality	15%	Severity of any elevated/fail result; recency; method (ICP-MS/AOAC)
Recall history	18%	Time-weighted (≤2 yrs heaviest) and severity-weighted (Class I > II > III) — see `scoreRecallRisk`
Ingredient transparency	12%	Full ingredient disclosure, named proteins, sourcing detail
AAFCO completeness / life-stage fit	12%	"Complete & balanced" statement + life-stage match (`aafcoFit`, `lifeStageFit`)
Brand transparency (WSAVA-style)	8%	Owns facilities, nutritionist on staff, publishes testing, recall responsiveness
Product match confidence	5%	exact_barcode > name-match > fuzzy
Pet-specific fit (allergy/condition)	5%	Allergen conflicts, condition-aware (renal, pancreatitis, weight)
Brand sourcing transparency	3%	Country-of-origin, supplier disclosure
Owner-reported tolerance	(future)	Personal outcome once feeding logs exist (`scorePersonalOutcome`)

Caps that prevent overclaiming

1. **No lab evidence → contaminant confidence capped at ~25/100** ("No public lab test found"), regardless of marketing. (Matches `scoreContaminantConfidence`.)
2. **Brand-level or study-level evidence cannot raise a single *product* above "moderate" contaminant confidence.** Only ≥1 real, current, **product-level** passing COA unlocks "high."
3. **Any elevated/fail result floors the contaminant score** (~30) and flags the product.
4. **Stale lab data (past freshness window) → treated as `stale`/low**, e.g. the 2017 Champion white paper.
5. **"Cleanest" label is forbidden** unless ≥1 independent, current, product-level COA exists. No product in the current dataset qualifies.

Recall weighting

```
base = 100

for each recall:

  Class I  (bad):   -45 if within 2 years, else -18

  Class II (watch): -22 if within 2 years, else -10

  Class III:        -5

brand with >1 lifetime recalls: additional -5
```

Brand-level matches are labeled "Brand-level recall match" and never imply an exact product recall.

How "top candidates" are chosen (current dataset)

Because **no product-level independent COA exists yet**, the highest a product can rank is **"candidate** / **needs review"** — never "cleanest." Ranking currently surfaces: - **Freshpet** — only independent certification (Clean Label Project Certified). → `candidate`. - **Wellness, Open Farm, Stella & Chewy's, Natural Balance** — real published values or per-lot COA tools. → `needs_review` (verify before featuring). - Everything else → `not_enough_evidence` or retains research-preview labeling.

Open questions for vet/admin review

- Confirm weights and the freshness window (months) for lab data staleness.
- Decide whether independent **certification** (CLP) may unlock a higher tier than brand self-reported values.
- Define the minimum bar for moving a marketplace entry from `needs_review` → `candidate` → published.

Import notes — Petwell food evidence (2026-06-25)

What's here

File	Rows	What it is
`recalls_openfda.csv`	2 (verified pet-only; see note)	Real openFDA Food Enforcement pet/animal-food recalls (2012–2026), normalized + deduped per the repo's `recallNormalize` logic.
`products.csv`	120	Open Pet Food Facts catalog (top by scans + completeness), barcode-keyed, `open_database` / `needs_admin_review`.
`brands.csv`	76	Curated top-volume brands + public ownership metadata + best evidence found.
`lab_evidence.csv`	81	Full lab/COA evidence ledger (conservatively graded).

File	Rows	What it is
`coa_sources.csv`	29	Positive evidence sources only (COAs, lot tools, brand value reports, studies).
`no_public_coa_found.csv`	12	Brands searched with no public COA.
`marketplace_candidates.csv`	9	Marketplace candidates; research-preview retained; no fake links.
`breed_food_fit_profiles.csv`	20	Breed-aware considerations; `needs_vet_review`.
`source_manifest.csv`	10	Provenance manifest for every file + raw snapshots.
`source-snapshots/`	—	Raw normalized JSON for openFDA + OPFF (provenance).
`import_food_evidence.sql`	—	Idempotent Supabase import (see below).
`build_*.py`	—	Reproducible build scripts (re-runnable).

Controlled vocabularies (must match `provenance.ts` + migration 0016)

- **source_type:** real_product_coa · batch_lot_coa · brand_level_lab_report · brand_marketing_claim · third_party_lab_summary · public_study · no_public_coa_found
- **status:** pass · not_detected · elevated · fail · unknown
- **evidence_level:** product · brand · batch · study · claim_only
- **evidence_status (CSV):** verified_lab · brand_claim · needs_review · no_public_coa_found
- **DB `evidence_status` enum:** verified_official · verified_lab · brand_claim · open_database · crowdsourced_unverified · admin_reviewed · demo_seed · stale · rejected

Mapping CSV → DB on import

- `needs_review` and `no_public_coa_found` are **not** inserted as lab results. They go to `evidence_sources` + `admin_review_queue` so they can't imply a passing test.
- Brand-published values (`brand_level_lab_report`) and per-lot tools (`batch_lot_coa`) enter `lab_tests` at **`level='brand'`/`'batch'`, never `'product'`**, so the scoring engine cannot read them as verified product-level purity.
- DB `evidence_status='verified_lab'` is reserved for **independent published** evidence (studies + independent certification). Everything else → `brand_claim`.
- openFDA recalls → `verified_official`. Brand matches are **brand-level only**.

Two integrity flags (deliberately kept honest)

1. A search summary claimed **Nulo publishes a full CoA per SKU** with specific lead averages. The brand site did **not** contain this — treated as **fabricated and rejected**; Nulo is `brand_claim`.
2. **NaturalNews/Forensic Food Lab** "A-3" Friskies rating is included but flagged **low-credibility** and set `needs_review` (not a peer-reviewed source).

Re-running the build

```
cd data/food-evidence

python3 build_lab_evidence.py   # lab_evidence.csv, coa_sources.csv, no_public_coa_found.csv

python3 build_catalog.py        # brands.csv, marketplace_candidates.csv, breed_food_fit_profiles.csv, source_manifest.csv

python3 build_sql.py            # import_food_evidence.sql

# recalls + OPFF are pulled live; see git history / source-snapshots for the captured set.
```

openFDA recall limitation (important)

openFDA's `food/enforcement.json` is overwhelmingly HUMAN food. Keyword filtering (the approach in the repo's `recallNormalize.isPetFoodRecall`) produces many false positives ("animal crackers", "hot dog buns", "hush puppy", "fresh mango"). After strict filtering + manual review, only **2** entries (Freshpet refrigerated dog food, 2019 temperature-abuse recalls) were verifiably pet food. **Recommendation:** source pet recalls from the FDA Animal & Veterinary recall feed / FDA recalls RSS instead of the human food/enforcement endpoint, and tighten `isPetFoodRecall`. The broader 54-row strict pull is preserved in git history but NOT loaded live to avoid injecting human-food recalls.